

Department of Management Studies
INDIAN INSTITUTE OF TECHNOLOGY MADRAS
MS5690
Computational Finance

Program: Research/ MBA

Core/Elective: Elective

Credits: 6

Period: Oct-Nov 2026

Pre-requisites: Basic knowledge of finance and statistics, along with introductory skills in data analysis or econometrics.

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COURSE DESCRIPTION

The aim of this course is to give an exposure on the basics of financial modeling and the use of statistical, econometric and nonlinear modeling in financial applications with practical case studies. Topics include the real-world implication of Auto-Regressive Integrated Moving Average (ARIMA), Generalized Autoregressive Conditional Heteroskedasticity (GARCH), Vector Autoregression (VAR), Vector Error Correction Model (VECM), Neural Network (NN) and Artificial Neural Network (ANN). This course provides students with a solid foundation in financial modeling and data analysis, equipping them with skills that are highly valued in finance and related fields. By understanding statistical, econometric and nonlinear modeling techniques, students will gain the ability to analyze complex financial data and interpret key insights. This course is instrumental for students aspiring to build careers in financial modeling, as it provides practical skills and technical knowledge essential for success in the field.

COURSE LEARNING OBJECTIVES (CLO)

CLO1: To develop a foundational understanding of financial modeling concepts, including different time series technique used in finance and their practical applications in real-world financial decision-making.

CLO2: To provide an understanding of data, their characteristics and preparing data for time series analysis and when and how to use techniques such as ARIMA and GARCH and how to interpret the result.

CLO3: To provide an understanding of when and how to use a structural model of data analysis and how to interpret the results.

CLO4: To develop practical expertise in academic-standard software such as EViews, SPSS, and STATA, learning how to use these tools to implement statistical and econometric models, enhance data analysis, and build complex financial models.

PEDAGOGY

The course is conducted through lectures, case study discussions/presentations and hands-on experience in developing financial models E Views, SPSS and STATA.

TEXTBOOK (S) AND READING MATERIALS

Recommended Text Books Textbook

Brooks, Chris, Introductory Econometrics for Finance, Cambridge University Press, Cambridge, 2008.

Reference books:

- Delurgio, Stephen, A. Forecasting Principles and Applications, Irwin McGraw Hill, Boston, 1998.
- Greene, William. H, Basics Econometrics, McGrawHill Inc.New York, 2000.
- Gujarati, Damodar, N., Basic Econometrics, McGrawHill, New York,1995.
- Alexander, Carol (ed)., Risk Management and analysis (Vol.1): Measuring and modeling Finance Risk, John Wiley and sons,1998.
- Stampfli, Joseph and Goodman, Victor, The Mathematics of Finance: Modeling and Hedging, Brooks/Cole, Thomson Learning, Pacific Groove, 2001.

TECHNOLOGY REQUIREMENTS AND SKILLS

- Basic of finance
- Participants are expected to have moderate skills of using spreadsheet
- Ability to use course management software Moodle
- Workflow software for course registration, add/drop, grades and reports

EXPECTATIONS AND GRADING

Students enrolled in this course must actively participate in discussion forums on case studies of different forecasting techniques. There are grades and assignments throughout the course with specific due dates.

Evaluation component	Weight	Description
Case Study	15%	Individual tasks involve students applying univariate and multivariate analysis techniques, such as t-tests for comparing group means, ANOVA and F-tests for examining variance among multiple groups, and descriptive statistics for summarizing data characteristics. These analyses is set in practical scenarios, enhancing students' ability to interpret financial data and make data-driven decisions.
Quiz (Individual)	15%	The quiz focus on foundational time series techniques, drawing from key concepts in a selected base paper to help students reinforce their understanding. Topics include essential methods such as ARIMA and GARCH, interpretation of results, and practical applications in financial forecasting. This approach ensures that students can apply theoretical knowledge to real-world scenarios, solidifying their grasp of time series analysis in financial contexts.
Mid-Term Project (individual)	30%	Students are required to select research papers that apply any five techniques, including ARIMA, GARCH, Regression, Simultaneous Equation Modeling, Vector Autoregression (VAR), Vector Error Correction Models (VECM), Neural Networks, and Panel Data Analysis. For each selected paper, students explain why the author chose the specific technique, linking it to the research objectives. Additionally, students describe the methodology used in the paper, highlighting how the technique is applied to address the research question(s).
End term project (individual)	40%	For the end-term project, students develop a financial model using daily stock data from NSE(Spot price, futures price, Option price, index, sectoral index, and relevant macroeconomic variable (from valid source)), focusing on univariate and structural models. The univariate model involves ARIMA and GARCH analysis, while the structural model use techniques like VAR/VECM and regression. Students forecast the out-sample period, evaluate model performance, and compare results with traditional forecasting methods. Guidelines for the doing the term paper is given separately.

TENTATIVE SESSION PLAN

Session	Date	Topic	Reading
1.	25/09/2026	Overview of Computational finance-Scope and applications-Choice of techniques-fundamental theories in finance and financial modeling – preparing data for modeling	Lecture & discussion
2.	06/10/2026	Testing properties of data for analysis-exercises	Using SPSS for preparing data – properties of data
3.	09/10/2026	Testing properties of data for analysis-exercises... Continuation	Using SPSS for preparing data – properties of data
4.	13/10/2026	Time series Analysis - ARIMA –Eviews application	Chapter 5 Research Paper: Forecasting daily returns of exchange rates using ARIMA and NN A hybrid probabilistic fuzzy ARIMA model for consumption forecasting in commodity markets https://doi.org/10.1016/j.eap.2017.12.003
5.	16/10/2026	Measuring time series volatility Eviews – using GARCH	Chapter 8 Research Paper: Impact of Index Derivatives on S&P CNX Nifty Volatility: Information Efficiency and Expiration Effects ESG indicators and sectoral volatility: GARCH vs. hybrid machine learning https://doi.org/10.1016/j.frl.2025.108103 Green bonds & clean energy in sustainable finance: Evidence from DCC-GARCH connectedness https://doi.org/10.1016/j.irfa.2025.104168
6.		Case Study	
7.	23/10/2026	Structural models - Applications using Multiple regression analysis	Chapter 2, 3, 4 Research Paper On the determinants of FDI and Portfolio Inflows: A cross-country study Effect of green inclusive finance on carbon emission efficiency: Empirical evidence from provincial panel data in China https://doi.org/10.1016/j.irfa.2025.104486
8.	27/10/2026	Structural models – Simultaneous Equation Modeling	Chapter 6 Research Paper: Do interbank markets price systemic risk? https://doi.org/10.1016/j.jfs.2024.101223
9.		Case Study	
10.	30/10/2026	Structural models – VAR-Eviews application	Chapter 6 Research Paper: Re-examining the nexus between Chinese carbon markets with energy and non-energy commodity markets in a novel risk spillover network approach

Session	Date	Topic	Reading
			https://doi.org/10.1016/j.pacfin.2025.102820
11.	03/11/2026	Structural models – Cointegration, VECM-Eviews/Stata application	Chapter 7 Research paper: Long run relationship between interest rates and inflation https://doi.org/10.1016/0378-4266(89)90031-9
12.	06/11/2026	Neural network application	Research paper: Guidelines for financial forecasting with Neural Networks
13.	10/11/2026	Panel Data Analysis	Research paper: Herding behaviour in P2P lending markets https://doi.org/10.1016/j.jempfin.2021.05.005
14.		Developing a Financial Model	
		End-Term Examination	

POLICIES

Attendance: Students must know the attendance policy of IIT Madras from the specific Ordinance of the program and meet the requirements. Ordinances of the programs of IIT Madras can be accessed at:

<https://www.iitm.ac.in/academics/academic-services/ordinances>

It is the student's responsibility to reach the classroom on time to ensure entry into the class for attendance.

Participants of this course are expected to practice and uphold standards of academic integrity and honesty. Academic integrity means representing oneself and one's work honestly. Misrepresentation is cheating since it means students are claiming credit for ideas or work not actually theirs and are thereby seeking a grade that is not actually earned. Following are some examples of academic dishonesty:

a. Cheating includes using materials such as books and/or notes when not authorized by the instructor, copying from someone else's paper, helping your peer to copy, theft of exam papers, falsifying data or submitting data not based on one's own work, and or other forms of misconduct on exams.

b. Plagiarizing the work of others. Plagiarism involves intellectual dishonesty. It is about using others' work without giving due credit to the others; by doing this, students are, in effect, claiming credit for someone else's thinking. Both direct quotations and paraphrases must be acknowledged through appropriate in-text/end referencing. Even if students rephrase, summarise or of select from others' work, ideas are still not theirs, and failure to give due credit constitutes misrepresentation, and plagiarism of another's work.

c. Unauthorized collaboration on computer assignments and unauthorized access to and use of computer programs, including modifying computer files created by others and representing that work as one's own.

d. Chat GPT is not allowed to do assignments in the course. Students are not directed to rely on Chat GPT or AI based chatbots for assignments, in-class discussions, etc.